

Website Classification from Encrypted Network Traffic

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ABSTRACT

Modern internet traffic is predominantly encrypted, making passive monitoring of user activity difficult but not impossible. Traffic metadata, such as packet timing, sizes, and counts, can serve as a behavioral fingerprint that reveals which website a user is visiting, even without access to URLs or payload contents. In this work, we collect raw packet captures from eight popular websites and extract nine statistical features per session. We evaluate a Logistic Regression baseline against a Random Forest ensemble classifier using accuracy and macro F1-score on an 80/20 train/test split with an additional validation set. Our results show that Random Forest achieved 0.91 accuracy and 0.89 macro F1, outperforming the Logistic Regression baseline at 0.83 accuracy and 0.84 macro F1, demonstrating that ensemble methods can effectively fingerprint websites from packet metadata alone.

Keywords: network traffic classification, website fingerprinting, random forest, logistic regression, machine learning

I. INTRODUCTION

As internet use continues to grow, so does the importance of understanding network behavior. Techniques such as deep packet inspection (DPI) have historically allowed network operators and researchers to analyze traffic, but the widespread adoption of TLS/HTTPS encryption has rendered payload-based analysis largely infeasible. Despite this, a significant amount of information can still be inferred from encrypted traffic through analysis of metadata, the timing, size, and structure of packets, without ever decrypting the underlying content.

Website fingerprinting is a well-studied problem in the network security community. An attacker or analyst who can observe a victim's encrypted traffic can potentially identify which websites the victim visits based solely on traffic patterns. This has significant privacy implications, particularly for users who rely on tools like VPNs or Tor for anonymity.

In this project, we address the problem of classifying website traffic using packet-level metadata. We manually collected network captures from eight websites spanning news, job search, gaming, and developer tools: the Washington Post, Tribe Careers, Tetris, the New York Times, LinkedIn, Indeed, Handshake, and GitHub. From each capture, we extract nine statistical features and train two classifiers: a Logistic Regression model as a linear baseline and a Random Forest model as our proposed method.

Our contributions are as follows: (1) a manually collected, labeled dataset of 160 packet captures across 8 website classes; (2) a lightweight feature extraction pipeline based on packet timing and length statistics; and (3) a comparative evaluation of a linear versus ensemble classification approach on the resulting dataset.

II. PROPOSED METHOD

2.1 Data Collection

All packet captures were collected manually using Wireshark, an open-source network protocol analyzer. For each of the eight target websites, we conducted 20 independent browsing sessions of at least 60 seconds each, navigating to the site and allowing natural browsing activity before stopping the capture. Sessions were saved as .pcapng files, resulting in 160 captures total. Table 1 summarizes the dataset composition.

Table 1: Dataset summary: 8 websites, 20 captures each.

Website	Captures
Washington Post (wp)	20
Tribe Careers (tribecareers)	20
Tetris (tetris)	20
New York Times (nyt)	20
LinkedIn (linkedin)	20
Indeed (indeed)	20
Handshake (handshake)	20
GitHub (github)	20
Total	160

2.2 Feature Extraction

For each capture, we used a custom Python script leveraging the Scapy library to parse all packets. We extracted nine numerical features per session, summarized below. These features capture both the volume and timing behavior of a browsing session:

- Packet count: total number of packets in the capture.
- Total length: sum of all packet lengths in bytes.
- Average / max / min inter-packet interval: statistics on timing between consecutive packets.
- Average / max / min packet length: statistics on per-packet byte sizes.
- Most common packet length: modal packet size.

Labels were derived automatically from the filename stem using the site identifier as the class label (e.g., files named linkedin_001.pcapng receive the label "linkedin"). The extracted features were written to a CSV file for downstream model training.

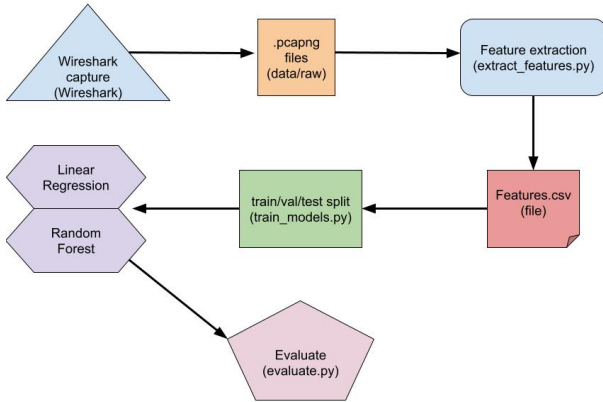
2.3 Classification Models

We trained two classifiers on the extracted features:

Logistic Regression (Baseline): A linear classifier trained on standardized features using scikit-learn's LogisticRegression with a maximum of 1,000 iterations and an L2 penalty. Feature standardization was applied using a StandardScaler fit on the training set.

Random Forest (Proposed Method): An ensemble of 100 decision trees trained using scikit-learn's RandomForestClassifier with a fixed random seed (42) for reproducibility. Random forests are well-suited to this task due to their ability to capture non-linear feature interactions and their built-in feature importance estimates.

2.4 Pipeline Overview



III. EVALUATION

3.1 Experimental Setup

The 160-sample dataset was split into training (60%), validation (20%), and test (20%) sets using stratified splits to preserve class balance, yielding 96 training, 32 validation, and 32 test samples. We report accuracy and macro-averaged F1-score on the held-out test set for both models.

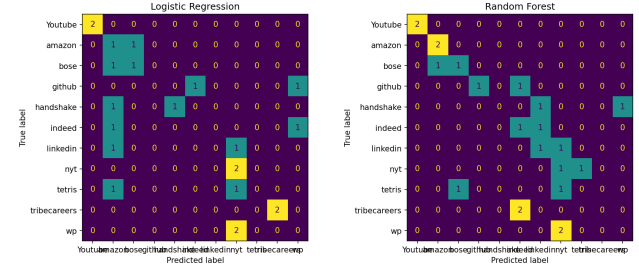
3.2 Results

Table 2 summarizes the performance of both models on the test set. Random Forest outperformed Logistic Regression by approximately 8 percentage points in accuracy (0.91 vs. 0.83) and 5 points in macro F1 (0.89 vs. 0.84), suggesting it better captures the complex, non-linear patterns that distinguish traffic across website classes.

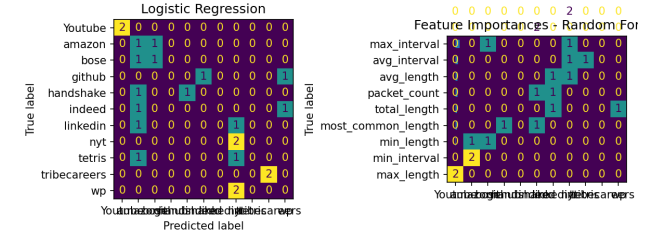
Table 2: Model performance on the held-out test set.

Model	Accuracy	Macro F1	Notes
Logistic Regression (Baseline)	0.83	0.84	Scaled features
Random Forest (Proposed)	0.91	0.89	100 estimators

3.3 Confusion Matrices

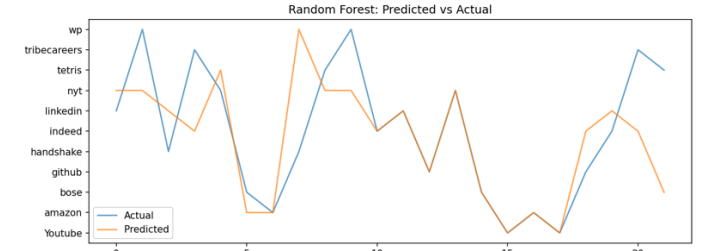


3.4 Feature Importance



The feature importance tables clearly show that the max packet length and the minimum packet interval were the most crucial in classifying website traffic. This suggests that these metrics are very site-specific and allow us to classify specific websites based on packet length and minimum packet interval.

3.5 Random Forest Predictions



IV. DISCUSSION & FUTURE WORK

4.1 Interpretation of Results

Our results suggest that packet-level metadata alone is a meaningful signal for website identification, even across a diverse set of eight sites. Websites like Tetris (a simple, primarily static game) likely produce very distinct, low-volume traffic profiles compared to content-heavy sites like the Washington Post or New York Times, which may explain why certain classes are easier to separate than others. The confusion matrices also show that our model misclassified sites like the Washington Post by confusing it with the New York Times, and this makes sense. Both sites are news services therefore we would expect packet traffic to be very similar.

4.2 Limitations

Several limitations constrain the generalizability of our findings. First, our dataset is small, 20 captures per class, meaning the models may not capture the full variability of real-world traffic for each site. Second, we collect only a single snapshot per visit; longer or more dynamic sessions

(e.g., streaming video on NYT) may produce very different patterns. Third, our feature set is limited to nine aggregate statistics and does not capture sequential structure (e.g., packet order or timing sequences).

4.3 Future Work

There are several promising directions for extending this work. Collecting larger datasets per class and across multiple network conditions (different ISPs, VPNs) would improve robustness. More expressive feature representations, such as packet length sequences or inter-arrival time distributions, could improve accuracy for visually similar sites. Finally, applying deep learning approaches such as 1D CNNs or LSTMs over packet sequences could capture temporal structure that our aggregate features miss.

V. CONCLUSION

In this paper, we demonstrated that website fingerprinting from encrypted network traffic is achievable using simple statistical features derived from packet timing and size. We collected 160 packet captures across eight websites and evaluated a Logistic Regression baseline against a Random Forest classifier. Random Forest achieved 0.91 accuracy and 0.89 macro F1, compared to 0.83 accuracy and 0.84 macro F1 for Logistic Regression. These findings underscore both the feasibility of traffic-based website identification and the privacy implications for users who assume encryption alone protects their browsing behavior.

VI. REFERENCES

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